



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

AUGSD/AGSRD

Course Handout

FIRST SEMESTER 2026-27

In addition to the part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : **MPBA G531**
Course Title : FINANCIAL DERIVATIVES AND ANALYSIS
Instructor-in-charge : Prof. Daitri Tiwary
Instructor(s) : Prof. Daitri Tiwary (daitri.tiwary@pilani.bits-pilani.ac.in)
Office hours for consultation : Prof. Daitri Tiwary: Monday 3PM to 4PM
Class timing: : Tuesday, Thursday & Friday, 12-12:50 PM

1. Scope of the Course:

This course introduces students to the fundamentals of financial derivatives and their applications in managing financial risk. It covers the basic concepts of forwards, futures, options, and swaps, along with their trading mechanisms, pricing principles, and hedging strategies. Students will also gain an introductory understanding of option pricing models, including the Binomial and Black–Scholes models, and explore the use of derivative instruments in equity, currency, interest rate, and fixed-income markets.

2. Course Objectives:

Upon successful completion of this course, students will be able:

1. **To explain** the fundamental concepts, features, and applications of financial derivatives, including forwards, futures, options, and swaps.
2. **To distinguish** between different derivative instruments and their use in managing market, currency, and interest rate risks.
3. **To apply** basic option trading, hedging, and pricing techniques to analyze financial risk management scenarios.
4. **To demonstrate** the use of introductory derivative pricing models, including the Binomial and Black–Scholes models, for valuing financial derivatives.
5. **To evaluate** the suitability of derivative instruments for hedging and investment decisions in equity, currency, interest rate, and fixed-income markets.

3. Text Book:

- T1. [An Introduction to Derivatives & Risk Management](#) by Don M Chance & Robert Brooks by Cengage Learning
- Options, Future & Other Derivatives by John C Hull and Sankarshan Basu 10th Edition, Prentice Hall

Reference:

- NISM Series VIII [Study Material](#)

4. Course Plan:

S. No.	Learning Objective	Topics to be covered	Ref. Ch. & Lecture No.
1.	Introduction to Derivatives	- Basics of Derivatives - Derivatives market – Evolution - Derivatives market – Instruments - Derivatives market – Participants	Class Notes Lecture. 1 to 3
2.	Important concepts in Financial Derivatives	- Short Selling - Repurchase Agreements - Risk & Return: Risk Preferences - Market Efficiency & Theoretical Fair Value - Risk Management: Case Study “Does The Use of Derivatives Impact Bank Performance? A Case Study of Relative Performance during 2002-2009”	Textbook1. Ch1 Lecture.4 to 5
3.	Futures and Forward Contracts	- The structure of Forward and Futures Market - Futures Arbitrage Strategies - Forward and Futures Hedging, Spread and Target Strategies - Swaps: Credit Defaults	Textbook1. Ch.8,10,11 & 12 Lecture.6 to 10
4.	Currency Derivatives	- Carry Trades and Foreign Currency Arbitrage - Currency Futures - Forward Contracts - Currency Swaps - Hedging Concepts - Hedging Contingent Foreign Currency Risk	Textbook1. Ch.10 & 11 Lecture.11 to 16
5.	Equity Index Futures	- Understanding Index - Attributes of an Index - Index Construction, Maintenance and Revision: Case of BSE Indices - Target Beta with Stock Index Futures	Class Notes Lecture. 17 to 20
6.	Bond Futures	- Treasury Bond Futures - Intermediate- and Long-Term Interest Rate Arbitrage - Target Duration with Bond Futures - Tactical Asset Allocation Using Bond Futures	Textbook1. Ch.10 & 11 Lecture. 21 to 24
7.	Introduction to Options	- Structure & Development of Options Market - Call & Put Options - Over-the-Counter Options - Options Terminology	Textbook.1 Ch.2 Lecture. 25 to 28
8.	Option pricing fundamentals	- Spot, Strike, Time Decay - Volatility - Option Greeks - Analysis of Call Option Trading from Buyer’s versus Seller’s Perspective - Analysis of Put Option Trading from Buyer’s versus Seller’s Perspective	Class Notes and Textbook.1 Ch.2 Lecture. 29 to 33
9.	Option Pricing Model	- The Binomial Pricing Model - The Black & Scholes Model - Using Black-Scholes-Merton Model to understand volatility	Textbook.1 Ch.4 & 5 Lecture. 34 to 36
10.	Option Strategies	- Terminology & Notations - Stock Transactions - Synthetic Puts and Calls - Option Spreads	Textbook.1 Ch.6 & 7 Lecture. 37 to 42

5. Evaluation Scheme:

The evaluation scheme consists of 60% closed-book components and 40% open-book components. Dates are indicative and may change subject to contingencies.

Component	Duration	Weightage (%)	Marks	Date & Time*	Nature of component (Close Book/ Open Book)
Quiz1 & Quiz2 (Best of two for evaluation)	20 Minutes	10	10	Quiz1 (Tuesday, August 18, 12 Noon) Quiz2 (Tuesday, September 22, 12 Noon)	Closed Book
Formulation and Presentation of Analytical Model Predicting Pricing of Financial Derivatives	15 Minutes	15	15	November 03, 05 & 06 2026	Open Book
Mid-Semester Examination	90 Minutes	25	25	06 th Oct (FN2: 11:00 A.M. to 12:30 P.M)	Open Book
Comprehensive Examination	180 Minutes	50	50	05 th Dec (AN: 3.00 P.M. to 6.00 P.M.)	Closed Book

*Slot timings are published and governed by AUGSD. Note if assessment components such as quiz/presentation dates coincide with holidays, the assessment will be postponed to the subsequent class.

6. Make-up Policy:

Consideration will be provided only for mid-term and comprehensive, given the proof of hospitalization and “*unfit for the test*” certificate from BITS medical center is submitted to the IC. The decision of IC will be final.

**Instructor-in-charge
(MPBA G531)**